into existence and then flash out in a random, unpredictable manner—here, there, and everywhere. Very occasionally a flashing wormhole will have one mouth near the ring world today and the other near Earth in the era 4 billion years ago when you embarked on your voyage. DAWN proposes to try to catch such a wormhole as it flickers, enlarge it like a child blowing up a balloon, and keep it open long enough for you to travel through it to the home of your youth.

But DAWN warns you of great danger. Physicists have conjectured, though it has never been proved, that an instant before an enlarging wormhole becomes a time machine, the wormhole must self-destruct with a gigantic, explosive flash. In this way the Universe might protect itself from time-travel paradoxes, such as a man going back in time and killing his mother before he was conceived, thereby preventing himself from being born and killing his mother.[46]

If the physicists' conjecture is wrong, then DAWN might be able to hold the wormhole open for a few seconds, with a large enough throat for you to travel through. By waiting nearby as she enlarges the wormhole and then plunging through it, within a fraction of a second of your own time you will arrive home on Earth, in the era of your youth 4 billion years ago. But if the time machine self-destructs, you will be destroyed with it. You decide to take the chance . . .

☆ ☆ ☆

The above tale sounds like science fiction. Indeed, part of it is: I cannot by any means guarantee that there exists a 10-solar-mass black hole near the star Vega, or a million-solar-mass hole at the center of the Milky Way, or a 15-trillion-solar-mass black hole anywhere at all in the Universe; they are all speculative but plausible fiction. Nor can I guarantee that humans will ever succeed in developing the technology for intergalactic travel, or even for interstellar travel, or for constructing ring worlds on girder-work structures around black holes. These are also speculative fiction.

On the other hand, I can guarantee with considerable but not complete confidence that black holes exist in our Universe and have the precise properties described in the above tale. If you hover in a blasting starship just above the horizon of a 15-trillion-solar-mass hole, I guarantee that the